Control

BaseController

```
# studentRepo: StudentRepository
# repRepo: CompanyRepRepository
# staffRepo: CareerCenterStaffRepository
# internshipRepo: InternshipRepository
# applicationRepo: ApplicationRepository
# withdrawalRepo: WithdrawalRequestRepository

# isEmailValid(email: String) : boolean
# isNullOrEmpty(str: String) : boolean
# validateNotNegative(value: int, fieldName: String) : void
```

CompanyRepRegistrationController

```
+ registerCompanyRep(name: String, companyName: String, department: String, position: String, email: String) : String
+ emailExists(email: String) : boolean
+ getPendingReps() : List<CompanyRep>
+ getRepByEmail(email: String) : CompanyRep
- validateRegistrationFields(name: String, companyName: String, department: String, position: String, email: String) : void
- isValidEmail(email: String) : boolean
```

CareerCenterStaffController

```
+ listPendingInternships() : List<Internship>
+ listUnapprovedReps() : List<CompanyRep>
+ listAllReps() : List<CompanyRep>
+ listPendingWithdrawals() : List<WithdrawalRequest>
+ listAllInternships() : List<Internship>
+ approveInternship(staffId: String, internshipId: String, makeVisible: boolean) : void
+ rejectInternship(staffId: String, internshipId: String) : void
+ approveCompanyRep(staffId: String, repEmail: String) : void
+ rejectCompanyRep(staffId: String, repEmail: String, reason: String) : void
+ approveWithdrawal(staffId: String, wrId: String, note: String) : void
+ rejectWithdrawal(staffId: String, wrId: String, note: String) : void
+ filterInternships(status: InternshipStatus, major: String, companyName: String, level: String) : List<Internship>
- validateStaffExists(staffId: String) : CareerCenterStaff
```

AuthController

```
- pw : Map<String, String>

+ seedDefaults(students: StudentRepository, reps: CompanyRepRepository, staff: CareerCenterStaffRepository) : void
+ loginLoginKey(loginKey: String, password: String) : String
+ changePassword(loginKey: String, oldPw: String, newPw: String) : void
+ verify(loginKey: String, password: String) : boolean
+ getUserRole(loginKey: String) : Role
- verifyAndSet(authKey: String, oldPw: String, newPw: String) : void
- requireReps() : void
- key(r: Role, id: String) : String
- normEmail(email: String) : String
```

<<enumeration>> Role

```
STUDENT
COMPANY_REP
STAFF
```

CompanyRepController

```
+ listInternships(repEmail: String) : List<Internship>
+ getInternship(repEmail: String, internshipId: String) : Internship
+ createInternship(repEmail: String, title: String, description: String, level: InternshipLevel, preferredMajor: String, openDate: LocalDate, closeDate: LocalDate, maxSlots: int) : String
+ editInternship(repEmail: String, internshipId: String, title: String, description: String, level: InternshipLevel, preferredMajor: String, openDate: LocalDate, closeDate: LocalDate, maxSlots: int) : void
+ deleteInternship(repEmail: String, internshipId: String) : void
+ setVisibility(repEmail: String, internshipId: String, visible: boolean) : void
+ listApplicants(repEmail: String, internshipId: String) : List<InternshipApplication>
+ markApplicationSuccessful(repEmail: String, applicationId: String) : void
+ markApplicationUnsuccessful(repEmail: String, applicationId: String) : void
- validateRepExists(repEmail: String) : CompanyRep
- validateOwnership(rep: CompanyRep, internship: Internship) : void
- validateInternshipParams(title: String, description: String, level: InternshipLevel, preferredMajor: String, openDate: LocalDate, closeDate: LocalDate, maxSlots: int) : void
```

StudentController

```
+ getStudent(studentId: String): Student
+ viewEligibleInternships(studentId: String): List<Internship>
+ viewFilteredInternships(studentId: String, major: String, level: InternshipLevel, companyName: String): List<Internship>
+ applyForInternship(studentId: String, internshipId: String): String
+ confirmAcceptance(studentId: String, applicationId: String): void
+ requestWithdrawal(studentId: String, applicationId: String, reason: String): String
+ viewMyApplications(studentId: String): List<InternshipApplication>
+ viewMyWithdrawalRequests(studentId: String): List<WithdrawalRequest>
+ getApplication(studentId: String, applicationId: String): InternshipApplication
+ canApplyMore(studentId: String): boolean
+ getActiveApplicationCount(studentId: String): int
+ hasConfirmedPlacement(studentId: String): boolean
+ getInternship(internshipId: String): Internship
```

DataSaveController

```
- baseDir : Path

+ saveAll(studentsCsv: String, repsCsv: String, staffCsv: String, internshipsCsv: String, applicationsCsv: String, withdrawalsCsv: String) : void
+ loadAll(studentsCsv: String, repsCsv: String, staffCsv: String, internshipsCsv: String, applicationsCsv: String, withdrawalsCsv: String) : void
+ getBaseDir() : Path
+ isInitialized() : boolean
```